

# Product Demand Prediction: AI Sprint Report

## 1. Project Overview

This report summarizes the development of a Multimodal Demand Prediction Engine. The system was engineered to forecast sales quantity (qty) by aligning visual design assets with historical transactional data from 'Sheet 2' of the provided dataset.

## 2. Technical Methodology

- Data Integrity: Processed 4 image ZIP files and matched 'code' columns with sales records.
- Vision Branch: CNN-based feature extraction to identify visual design patterns.
- Numerical Branch: Analysis of product 'rate' and historical demand trends.
- Vector Engine: FAISS implementation using Cosine Similarity for Strategy B (Enhanced Retrieval).

## 3. Data Insights & Results

Note: Dashboard visualization 'demand\_prediction\_custom.png' was not found during export.

## 4. Vertex AI Production Path

The system is architected for production deployment on Google Cloud Vertex AI. Future scaling includes utilizing Vertex AI Vector Search for global deployments and automated monitoring for model accuracy drift.